

AMAT 19

Finite Mathematics

Rules of Inference and Equivalence Laws of Tautological Forms



Objectives

1. Define the converse, contrapositive, and inverse of a conditional statement;
2. Identify some rules of inference and laws of equivalence of tautological forms;
3. Use truth table to prove logical implication and equivalence;
4. Use the rules of inference and equivalence laws to simplify statements.



Definition: Converse, Contrapositive and Inverse

Definition: Converse, Contrapositive and Inverse

Given the conditional statement $p \rightarrow q$.

<i>Converse</i>	$q \rightarrow p$
<i>Contrapositive</i>	$\sim q \rightarrow \sim p$
<i>Inverse</i>	$\sim p \rightarrow \sim q$

Definition: Converse, Contrapositive and Inverse

Definition: Converse, Contrapositive and Inverse

Given the conditional statement $p \rightarrow q$.

<i>Converse</i>	$q \rightarrow p$
<i>Contrapositive</i>	$\sim q \rightarrow \sim p$
<i>Inverse</i>	$\sim p \rightarrow \sim q$

Example: Given the conditional statement below, give its converse, contrapositive and inverse.

If Doods gets perfect scores on her long exams, then she will get 1.0.

Logical Implication (or Rule of Inference)

A compound proposition P logically implies a compound proposition Q , written $P \Rightarrow Q$, whenever the conditional statement $P \rightarrow Q$ is a tautology.

Typically, a rule of inference preserves the truth.



Some Rules of Inference

Name	Implication
<i>Addition</i>	$p \Rightarrow (p \vee q)$
<i>Simplification</i>	$(p \wedge q) \Rightarrow p$
<i>Modus ponens</i>	$[p \wedge (p \rightarrow q)] \Rightarrow q$
<i>Modus tollens</i>	$[(p \rightarrow q) \wedge \sim q] \Rightarrow \sim p$
<i>Disjunctive Syllogism</i>	$[(p \vee q) \wedge \sim p] \Rightarrow q$
<i>Hypothetical syllogism</i>	$[(p \rightarrow q) \wedge (q \rightarrow r)] \Rightarrow (p \rightarrow r)$
<i>Constructive Dilemma</i>	$[(p \rightarrow q) \wedge (r \rightarrow s)] \wedge (p \vee r) \Rightarrow (q \vee s)$
<i>Distractive Dilemma</i>	$(p \rightarrow q) \Rightarrow [p \rightarrow (p \wedge q)]$

Logical Implication

Example: Prove that Addition is a Logical Implication

In order to prove that Addition $[p \Rightarrow (p \vee q)]$ is indeed a logical implication, we need to show that $p \rightarrow (p \vee q)$ is a Tautology. We use a truth table to establish this proof.

p	q	$p \vee q$	$p \rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

As an exercise, use truth tables to prove that the rules of inference shown in the previous slide are indeed a tautology.

Logical Equivalence

Compound propositions P and Q are logically equivalent, written $P \equiv Q$, $P \Leftrightarrow Q$, or P iff Q , if they have the same truth values for all possible truth values of their variables.

A logical equivalence that is frequently used is sometimes called a **logical identity**.

Facts:

1. $P \Leftrightarrow Q$ is true if and only if $P \Rightarrow Q$ and $Q \Rightarrow P$.
2. $P \Leftrightarrow Q$ is true if and only if $P \leftrightarrow Q$ is a tautology.

Exercise: Logical Equivalence

Which of the following statements is equivalent to “If it is a bird, then it flies.”? [Hint: Use truth tables to show equivalence.]

1. If it flies, then it is a bird.
2. If it is not a bird, then it does not fly.
3. If it does not fly, it is not a bird.

Exercise: Logical Equivalence

Which of the following statements is equivalent to “If it is a bird, then it flies.”? [Hint: Use truth tables to show equivalence.]

1. If it flies, then it is a bird. (converse)
2. If it is not a bird, then it does not fly. (inverse)
3. If it does not fly, it is not a bird. (contrapositive)

Exercise: Logical Equivalence

Which of the following statements is equivalent to “If it is a bird, then it flies.”? [Hint: Use truth tables to show equivalence.]

1. If it flies, then it is a bird. (converse)
2. If it is not a bird, then it does not fly. (inverse)
3. If it does not fly, it is not a bird. (contrapositive)

Remark:

Their truth tables should show that the given conditional statement is equivalent to its contrapositive, but not equivalent to its converse and its inverse. However, the converse and inverse of the conditional are equivalent statements.

Some Logical Equivalence

Name		Equivalence
<i>Double Negation</i>		$\sim\sim p \Leftrightarrow p$
<i>Commutative Laws</i>		$(p \vee q) \Leftrightarrow (q \vee p)$ $(p \wedge q) \Leftrightarrow (q \wedge p)$ $(p \leftrightarrow q) \Leftrightarrow (q \leftrightarrow p)$
<i>Associative Laws</i>		$[p \wedge (q \wedge r)] \Leftrightarrow [(p \wedge q) \wedge r]$ $[p \vee (q \vee r)] \Leftrightarrow [(p \vee q) \vee r]$
<i>Distributive Laws</i>		$[p \vee (q \wedge r)] \Leftrightarrow [(p \vee q) \wedge (p \vee r)]$ $[p \wedge (q \vee r)] \Leftrightarrow [(p \wedge q) \vee (p \wedge r)]$
<i>Idempotent Laws</i>		$(p \vee p) \Leftrightarrow p$ $(p \wedge p) \Leftrightarrow p$
<i>Identity Laws</i>		$(p \vee F) \Leftrightarrow p$ $(p \wedge T) \Leftrightarrow p$

Some Logical Equivalence (Cont.)

Name	Equivalence
<i>Domination Laws</i>	$(p \vee T) \Leftrightarrow T$ $(p \wedge F) \Leftrightarrow F$
<i>Negation Laws</i>	$(p \vee \sim p) \Leftrightarrow T$ $(p \wedge \sim p) \Leftrightarrow F$
<i>De Morgan's Laws</i>	$\sim (p \vee q) \Leftrightarrow (\sim p \wedge \sim q)$ $\sim (p \wedge q) \Leftrightarrow (\sim p \vee \sim q)$
<i>Contrapositive</i>	$(p \rightarrow q) \Leftrightarrow (\sim q \rightarrow \sim p)$
<i>Implication</i>	$(p \rightarrow q) \Leftrightarrow (\sim p \vee q)$ $(p \rightarrow q) \Leftrightarrow \sim (p \wedge \sim q)$
<i>Equivalence</i>	$(p \leftrightarrow q) \Leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow p)]$

Logical Equivalence

Example: Prove that Double Negation is a Logical Equivalence

In order to prove that Double Negation $[\sim\sim p \Leftrightarrow p]$ is indeed a logical equivalence, we need to show that $\sim\sim p \leftrightarrow p$ is a Tautology (or the left and right propositions have the same truth values for each case). Its truth table establishes this proof.

p	$\sim p$	$\sim(\sim p)$
T	<i>F</i>	T
F	<i>T</i>	F

As an exercise, use truth tables to prove that the logical equivalence shown on the two previous slides are indeed a tautology.

Exercise 1

Eliminate the connective $\rightarrow$ in the compound proposition and simplify as much as possible:

$$(\sim p \wedge \sim q) \rightarrow (q \rightarrow r).$$

Proof.

You may use any of the logical implication rules and logical equivalence laws to eliminate the connective and simplify the statement. The rules of inference may be used to reduce statements, however, this may not be used for proving equivalence of statements.

Mathematical Statement	Reason
$(\sim p \wedge \sim q) \rightarrow (q \rightarrow r)$	Given
$(\sim p \wedge \sim q) \rightarrow (\sim q \vee r)$	Implication
$\sim (\sim p \wedge \sim q) \vee (\sim q \vee r)$	Implication
$(\sim \sim p \vee \sim \sim q) \vee (\sim q \vee r)$	De Morgan's Law
$(p \vee q) \vee (\sim q \vee r)$	Double Negation
$p \vee (q \vee \sim q) \vee r$	Associativity of Disjunction
$p \vee (\mathbf{T}) \vee r$	Negation Laws
$(p \vee \mathbf{T}) \vee r$	Associativity of Disjunction or Precedence of Order of Operations
$\mathbf{T} \vee r$	Domination Laws
$\mathbf{T}$	Domination Laws

□

Here, we have shown that the statement $(\sim p \wedge \sim q) \rightarrow (q \rightarrow r)$ is reduced to a **True** value. You may also try to check if this is indeed correct by using the truth table. You should end up with a **Tautology**.

Exercise 2

Rewrite the compound proposition using only the connectives $\sim$ and $\vee$:

$$(p \rightarrow q) \wedge (q \vee r).$$

Proof.

Similar to Exercise 1, since we are only tasked to rewrite the compound propositions to a statement involving the connectives $\sim$ and $\vee$, (and not proving equivalence of statements), we may use both rules of inference and laws of equivalence of tautological forms. You may end up with your own statement in the end as long as it only involves the connectives $\sim$ and $\vee$. Here is one way to rewrite the compound statement. It is important to always state the reason after transformation to understand why it arrived to such statement.

Mathematical Statement	Reason
$(p \rightarrow q) \wedge (q \vee r)$	Given
$(\sim p \vee q) \wedge (q \vee r)$	Implication
$(q \vee \sim p) \wedge (q \vee r)$	Commutativity of Disjunction
$q \vee (\sim p \wedge r)$	Distributive Laws
$q \vee \sim (p \vee \sim r)$	De Morgan's Laws

□

At this point, we have reduced the compound statement to a statement that contains only the connectives $\sim$ and $\vee$. What did you get?